

COSC 594

LLM Composition

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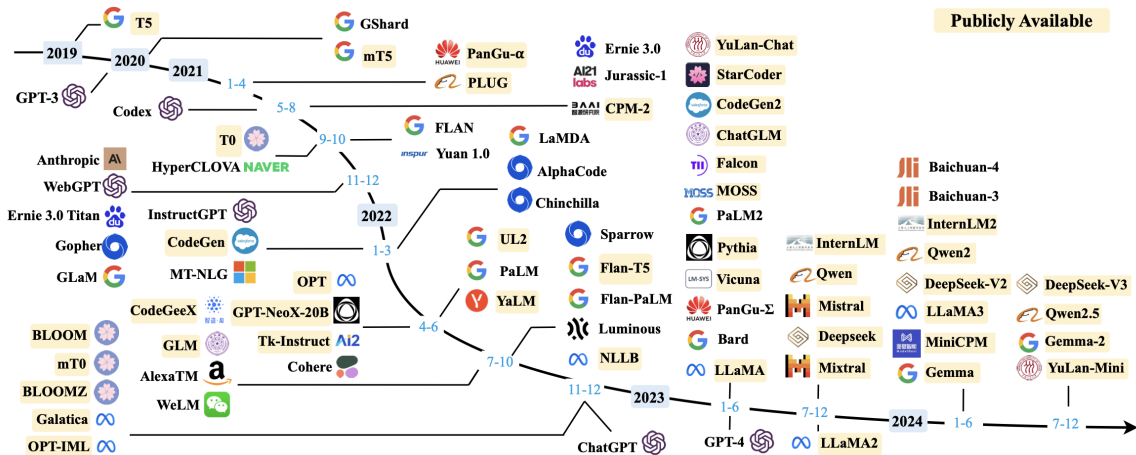
Outline

- MOTIVATION
- EXAMPLES
- REGULARIZATION-BASED APPROACH
- CONSTRAINED APPROACH

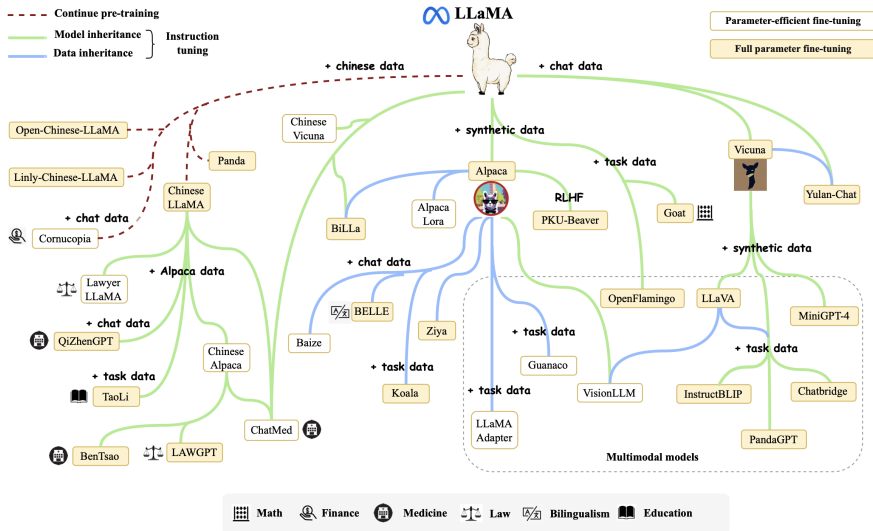
Outline

■ MOTIVATION

The growth of trained LLMs



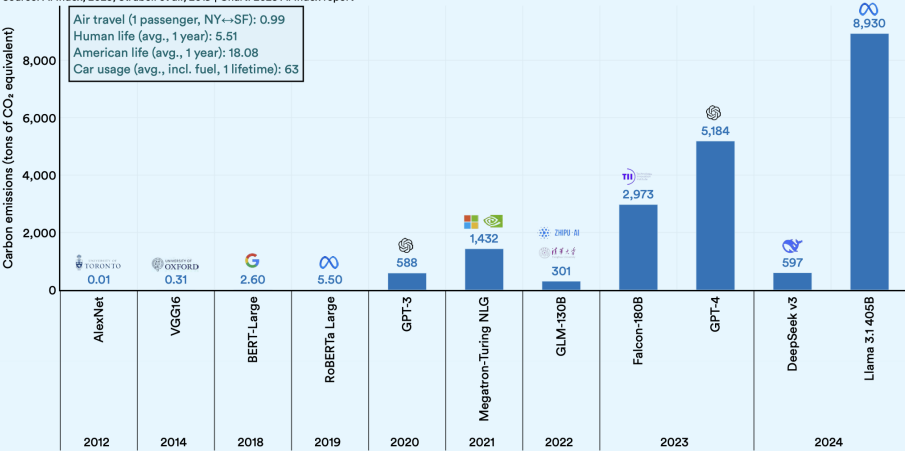
A growing number of variant LLMs



Carbon footprint

Estimated carbon emissions from training select AI models and real-life activities, 2012–24

Source: AI Index, 2025; Strubell et al., 2019 | Chart: 2025 AI Index report



Climate change

image source: The 2025 AI Index Report

Generalization

A → B



Who is **Tom Cruise's** mother?



Tom Cruise's mother is **Mary Lee Pfeiffer**.



B → A



Who is **Mary Lee Pfeiffer's** son?



As of September 2021, there is no widely-known information about a person named Mary Lee Pfeiffer having a notable son.

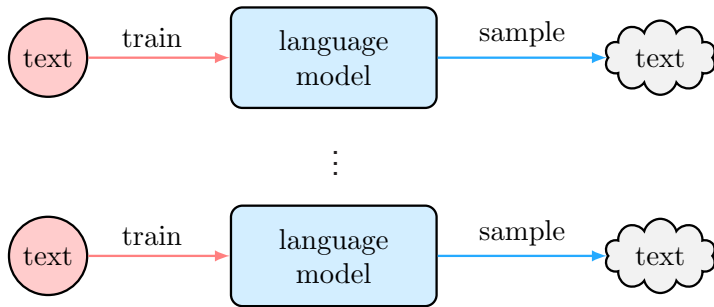


Reversal curse

Combination of LLMs



⋮



$$y \sim \pi_0(\cdot | x), \pi_1(\cdot | x), \dots, \pi_K(\cdot | x)$$

How do we combine several pretrained LLMs?

Outline

■ MOTIVATION

■ EXAMPLES

Model merging

$K + 1$ **trained models** : $\pi_{\theta_0}, \pi_{\theta_1}, \dots, \pi_{\theta_K}$

■ WEIGHTED AVERAGING

$$\text{merge}(\theta_0, \theta_1, \dots, \theta_K) = \frac{1}{K+1} \sum_{k=0}^K w_k \theta_k$$

■ TASK ARITHMETIC

$$\text{merge}(\theta_0, \theta_1, \dots, \theta_K) = \frac{1}{K} \sum_{k=1}^K t_k \tau_k + \theta_0$$

$\tau_k := \theta_k - \theta_0$ – k th task

■ MIXTURE OF EXPERTS

$$\text{merge}(\pi_{\theta_0}, \pi_{\theta_1}, \dots, \pi_{\theta_K}) = \frac{1}{K+1} \sum_{k=0}^K \lambda_k \pi_{\theta_k}$$

Heuristic design of (w_k, t_k, λ_k) & performance varies

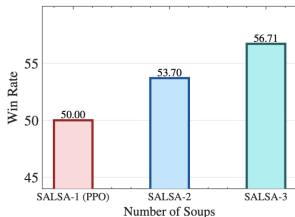
Soup-based alignment

Soup model: $\pi_{\text{soup}} = \pi_{\frac{1}{K+1} \sum_{k=0}^K w_k \theta_k}$

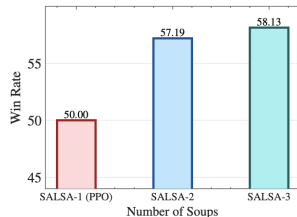
$$\log \pi_{\text{soup}} = \sum_{k=0}^K w_k \log \pi_{\theta_k}$$

■ PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi_{\theta}(\cdot | x) \| \pi_{\text{soup}}(\cdot | x)) \right]$$



(a) Arena-Hard



(b) MT-Bench

Outline

- MOTIVATION
- EXAMPLES
- REGULARIZATION-BASED APPROACH

Mixture of regularizations

RLHF w/ many reference models

$K + 1$ trained models : $\pi_0, \pi_1, \dots, \pi_K$

■ PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_\theta(\cdot | x)} [r(x, y)] - \sum_{k=0}^K \beta_k D_{\text{KL}}(\pi_\theta(\cdot | x) \parallel \pi_k(\cdot | x)) \right]$$

$\beta_0, \beta_1, \dots, \beta_K$ – tradeoff parameters

- ★ k th policy π_k is pretrained
- ★ reward model r is from reward modeling

Alignment w/ **multiple reference models**

Policy gradient

■ REINFORCE

$$\begin{aligned} & \nabla_{\theta} \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [r(x, y)] - \sum_{k=0}^K \beta_k D_{\text{KL}}(\pi_{\theta}(\cdot | x) \parallel \pi_k(\cdot | x)) \right] \\ &= \nabla_{\theta} \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} \left[r(x, y) - \sum_{k=0}^K \beta_k \log \frac{\pi_{\theta}(y | x)}{\pi_k(y | x)} \right] \right] \\ &= \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} \left[(\tilde{r}(x, y) - \sum_{k=0}^K \beta_k) \nabla_{\theta} \log \pi_{\theta}(y | x) \right] \right] \end{aligned}$$

$$\tilde{r}(x, y) = r(x, y) - \sum_{k=0}^K \beta_k \log \frac{\pi_{\theta}(y | x)}{\pi_k(y | x)} - \text{regularized reward}$$

- ★ on-policy unbiased estimate
- ★ off-policy unbiased estimate

$$\mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\text{off}}(\cdot | x)} \left[\frac{\pi_{\theta}(y | x)}{\pi_{\text{off}}(\cdot | x)} \left(\tilde{r}(x, y) - \sum_{k=0}^K \beta_k \right) \nabla_{\theta} \log \pi_{\theta}(y | x) \right] \right]$$

Multi-reference RLHF in distribution space

■ DISTRIBUTION OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \sum_{k=0}^K \beta_k D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_k(\cdot | x)) \right]$$

$\beta_0, \beta_1, \dots, \beta_K$ – tradeoff parameters

★ optimal LLM policy $\pi^\star$

$$\pi^\star(y | x) = \pi_{\text{ref}}(y | x) e^{r(x, y) / \mathbf{1}^\top \beta} \frac{1}{Z(x)}$$

★ reference policy π_{ref}

$$\pi_{\text{ref}}(y | x) = \prod_{k=0}^K (\pi_k(y | x))^{\beta_k / \mathbf{1}^\top \beta} \frac{1}{Z_{\text{ref}}(x)}$$

$Z(x), Z_{\text{ref}}(x)$ – normalization constants

■ MULTI-REFERENCE KL DIVERGENCE

$$\begin{aligned}\sum_{k=0}^K \beta_k D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_k(\cdot | x)) &= \sum_{k=0}^K \beta_k \mathbb{E}_{y \sim \pi(\cdot | x)} [\log \pi(y | x) - \log \pi_k(y | x)] \\&= \mathbf{1}^\top \beta \mathbb{E}_{y \sim \pi(\cdot | x)} [\log \pi(y | x)] \\&\quad - \mathbf{1}^\top \beta \mathbb{E}_{y \sim \pi(\cdot | x)} \left[\log \prod_{k=0}^K (\pi_k(y | x))^{\beta_k / \mathbf{1}^\top \beta} \right] \\&= \mathbf{1}^\top \beta \mathbb{E}_{y \sim \pi(\cdot | x)} [\log \pi(y | x)] \\&\quad - \mathbf{1}^\top \beta \mathbb{E}_{y \sim \pi(\cdot | x)} [\log \pi_{\text{ref}}(y | x)] - \mathbf{1}^\top \beta \log Z_{\text{ref}}(x) \\&= \mathbf{1}^\top \beta D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_{\text{ref}}(\cdot | x)) - \underbrace{\mathbf{1}^\top \beta \log Z_{\text{ref}}(x)}_{\pi\text{-independent}}\end{aligned}$$

■ SINGLE-REFERENCE RLHF

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \sum_{k=0}^K \beta_k D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_k(\cdot | x)) \right]$$

$$\Updownarrow$$

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \mathbf{1}^\top \beta D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_{\text{ref}}(\cdot | x)) \right]$$

$\beta_0, \beta_1, \dots, \beta_K$ – tradeoff parameters

★ reference policy π_{ref}

$$\pi_{\text{ref}}(y | x) = \prod_{k=0}^K (\pi_k(y | x))^{\beta_k / \mathbf{1}^\top \beta} \frac{1}{Z_{\text{ref}}(x)}$$

$Z_{\text{ref}}(x)$ – normalization constant

Correspondence between policy and reward

$$\pi^*(y | x) = \prod_{k=0}^K (\pi_k(y | x))^{\beta_k / \mathbf{1}^\top \beta} e^{r(x,y) / \mathbf{1}^\top \beta} \frac{1}{Z(x) Z_{\text{ref}}(x)}$$

- ★ from reward to optimal LLM policy
- ★ from optimal LLM policy to reward

$$r(x, y) = \sum_{k=0}^K \beta_k \log \frac{\pi^*(y | x)}{\pi_k(y | x)} + \mathbf{1}^\top \beta \log(Z(x) Z_{\text{ref}}(x))$$

Rich reward function class

Reward representation trick

Reward decomposition: $r(x, y) = \sum_{k=0}^K r_k(x, y)$

Data set : $D = \cup_{k=0}^K \{x^{(i,k)}, y_1^{(i,k)}, y_0^{(i,k)}\}_{i=1}^N$

Reward modeling: $\text{maximize}_r \mathbb{E}_{(x, y_1, y_0) \sim D} [\log \mathbb{P}(y_1 \succ y_0 | x)]$

$$r_k(x, y) := \beta_k \log \frac{\pi^*(y | x)}{\pi_k(y | x)} \quad - \text{reward parametrization}$$

■ REWARD REPRESENTATION

$$\begin{aligned} & \mathbb{E}_{(x, y_1, y_0) \sim D} [\log \mathbb{P}(y_1 \succ y_0 | x)] \\ &= \sum_{k=0}^K \mathbb{E}_{(x, y_1, y_0) \sim D_k} [\log \sigma(r_k(x, y_1) - r_k(x, y_0))] \\ &= \sum_{k=0}^K \mathbb{E}_{(x, y_1, y_0) \sim D_k} \left[\log \sigma \left(\beta_k \log \frac{\pi^*(y_1 | x)}{\pi_k(y_1 | x)} - \beta_k \log \frac{\pi^*(y_0 | x)}{\pi_k(y_0 | x)} \right) \right] \end{aligned}$$

Multi-reference RLHF in parameter space

■ PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad \sum_{k=0}^K \mathbb{E}_{(x, y_1, y_0) \sim D_k} \left[\log \sigma \left(\beta_k \log \frac{\pi_{\theta}(y_1 | x)}{\pi_k(y_1 | x)} - \beta_k \log \frac{\pi_{\theta}(y_0 | x)}{\pi_k(y_0 | x)} \right) \right]$$

★ nonconvex parameter optimization

■ POLICY GRADIENT (DIRECT PREFERENCE OPTIMIZATION, DPO)

$$\begin{aligned} & \sum_{k=0}^K \nabla_{\theta} \mathbb{E}_{(x, y_1, y_0) \sim D_k} \left[\log \sigma \left(\beta_k \log \frac{\pi_{\theta}(y_1 | x)}{\pi_k(y_1 | x)} - \beta_k \log \frac{\pi_{\theta}(y_0 | x)}{\pi_k(y_0 | x)} \right) \right] \\ &= \sum_{k=0}^K \mathbb{E}_{(x, y_1, y_0) \sim D_k} \left[\sigma(\hat{r}_k(x, y_0) - \hat{r}_k(x, y_1)) (\beta_k \nabla_{\theta} \log \pi_{\theta}(y_1 | x) - \beta_k \nabla_{\theta} \log \pi_{\theta}(y_0 | x)) \right] \end{aligned}$$

$$\hat{r}(x, y) := \beta_k \log \frac{\pi_{\theta}(y | x)}{\pi_k(y | x)} \quad - \quad \text{reward parametrization}$$

Decentralized DPO

Personalized regularization

Multi-objective RLHF w/ many reference models

■ MULTI-OBJECTIVE PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \left\{ \underbrace{\mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_{\theta}(\cdot | x)} [r(x, y)] - \beta_k D_{\text{KL}}(\pi_{\theta}(\cdot | x) \parallel \pi_k(\cdot | x)) \right]}_{:= F^{(k)}(\theta)} \right\}_{k=0}^K$$

π_k – k th trained model

$\beta_k > 0$ – tradeoff parameter

★ Pareto solution θ^*

$$\neg \exists \theta \ F^{(k)}(\theta) \geq F^{(k)}(\theta^*) \text{ for all } k, \text{ and } F^{(k')}(\theta) > F^{(k')}(\theta^*) \text{ for some } k'$$

Intractability of Pareto solution

Scalarization approach

■ SINGLE-OBJECTIVE DISTRIBUTION OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \sum_{k=0}^K w_k \beta_k D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_k(\cdot | x)) \right]$$

π_k – k th trained model

$\beta_k > 0$ – tradeoff parameter

$\sum_{k=0}^K w_k = 1, w_k > 0$ – mixture weights

- ★ sufficient for multi-objective strictly convex optimization
- ★ Pareto LLM policy π^*

$$\pi^*(y | x) = \prod_{k=0}^K (\pi_k(y | x))^{\bar{\beta}_k / \mathbf{1}^\top \bar{\beta}} e^{r(x, y) / \mathbf{1}^\top \bar{\beta}} \frac{1}{Z(x) Z_{\text{ref}}(x)}$$

$$\bar{\beta}_k := w_k \beta_k$$

■ INFERENCE-TIME ALIGNMENT

$$\begin{aligned}\pi^\star(y \mid x) &= \prod_{k=0}^K (\pi_k(y \mid x))^{\bar{\beta}_k / \mathbf{1}^\top \bar{\beta}} e^{r(x,y) / \mathbf{1}^\top \bar{\beta}} \frac{1}{Z(x) Z_{\text{ref}}(x)} \\ &= \prod_{k=0}^K (\pi_k(y \mid x) e^{\bar{r}_k(x,y)})^{\bar{\beta}_k / \mathbf{1}^\top \bar{\beta}} \frac{1}{Z(x) Z_{\text{ref}}(x)} \\ &= \prod_{k=0}^K (\pi_k^\star(y \mid x))^{\bar{\beta}_k / \mathbf{1}^\top \bar{\beta}}\end{aligned}$$

$\bar{r}_k(x, y) := r(x, y) / \bar{\beta}_k$ – rescaled reward

$\Downarrow$

$$\log \pi^\star(y \mid x) = \sum_{k=0}^K \frac{\bar{\beta}_k}{\mathbf{1}^\top \bar{\beta}} \log \pi_k^\star(y \mid x)$$

Soup model

Worst-case regularization

Minimax multi-objective RLHF w/ many reference models

■ SINGLE-OBJECTIVE DISTRIBUTION OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{x,y \sim \pi(\cdot|x)} [r(x,y)] - \beta \underbrace{\underset{k}{\text{maximize}} \quad \mathbb{E}_x [D_{\text{KL}}(\pi(\cdot|x) \parallel \pi_k(\cdot|x))]}_{\text{discrete optimization}}$$

π_k – k th trained model

$\beta > 0$ – tradeoff parameter

■ CONSTRAINED DISTRIBUTION OPTIMIZATION

$$\underset{\pi, t}{\text{maximize}} \quad \mathbb{E}_{x,y \sim \pi(\cdot|x)} [r(x,y)] - \beta t$$

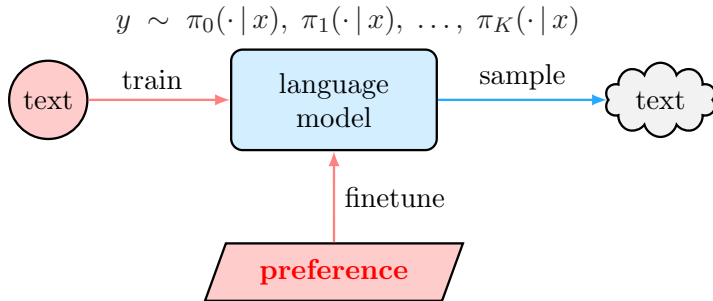
$$\text{subject to} \quad \mathbb{E}_x [D_{\text{KL}}(\pi(\cdot|x) \parallel \pi_k(\cdot|x))] \leq t \quad \text{for } k = 0, 1, \dots, K$$

★ linear objective and convex constraints

Outline

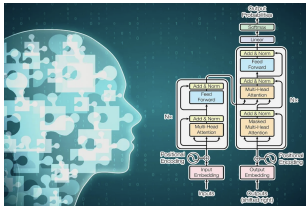
- MOTIVATION
- EXAMPLES
- REGULARIZATION-BASED APPROACH
- CONSTRAINED APPROACH

Combination of LLMs via constraints



preference – reward / human feedback

Text generation (ChatGPT)



Goal helpful responses

Requirements retain model #1

retain model #2

⋮

Constrained RLHF w/ many reference models

$K + 1$ trained models : $\pi_0, \pi_1, \dots, \pi_K$

■ CONSTRAINED PARAMETER OPTIMIZATION

$$\underset{\theta}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi_\theta(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi_\theta(\cdot | x) \parallel \pi_0(\cdot | x)) \right]$$

$$\text{subject to} \quad \mathbb{E}_x [D_{\text{KL}}(\pi_\theta(\cdot | x) \parallel \pi_k(\cdot | x))] \leq \epsilon_k$$

for $k = 1, \dots, K$

★ nonconvex objective and nonconvex constraints

Intractability of nonconvex constrained optimization

Constrained RLHF in distribution space

■ CONSTRAINED DISTRIBUTION OPTIMIZATION

$$\begin{aligned} & \underset{\pi}{\text{maximize}} \quad \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \beta D_{\text{KL}}(\pi_{\theta}(\cdot | x) \parallel \pi_0(\cdot | x)) \right] \\ & \text{subject to} \quad \mathbb{E}_x \left[D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_k(\cdot | x)) \right] \leq \epsilon_k \end{aligned}$$

for $k = 1, \dots, K$

★ concave objective function and convex constraints

High-dimensional, convex $\implies$ strong duality

Lagrangian approach

■ LAGRANGIAN

$$L(\pi; \lambda) = \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \sum_{k=0}^K \lambda_k D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_k(\cdot | x)) \right] + \lambda^\top \epsilon$$

$$\lambda_0 = \beta, \quad \lambda^\top \epsilon := \sum_{k=1}^K \lambda_k \epsilon_k$$

■ SINGLE-OBJECTIVE ALIGNMENT

$$\underset{\pi}{\text{maximize}} \quad L(\pi; \lambda)$$

★ explicit maximizer $\pi^*(\lambda)$

Unknown optimal dual variable λ^*

Strong duality

Dual problem: $\underset{\lambda}{\text{minimize}} \underset{\pi}{\text{maximize}} L(\pi; \lambda)$

Primal problem: $\underset{\pi}{\text{maximize}} \underset{\lambda}{\text{minimize}} L(\pi; \lambda)$

■ CONVEX-CONCAVE MIN-MAX OPTIMIZATION

$$\underset{\lambda}{\text{minimize}} \underset{\pi}{\text{maximize}} L(\pi; \lambda) = \underset{\pi}{\text{maximize}} \underset{\lambda}{\text{minimize}} L(\pi; \lambda)$$

Strong duality $\implies$ **exchange** of min and max

OBJECTIVE: find a minimax point (π^*, λ^*)

Dual problem

$$\underset{\lambda}{\text{minimize}} \quad \underset{\pi}{\text{maximize}} \quad L(\pi; \lambda)$$

$$L(\pi; \lambda) := \mathbb{E}_x \left[\mathbb{E}_{y \sim \pi(\cdot | x)} [r(x, y)] - \sum_{k=0}^K \lambda_k D_{\text{KL}}(\pi(\cdot | x) \parallel \pi_k(\cdot | x)) \right] + \lambda^\top \epsilon$$

■ SINGLE-OBJECTIVE ALIGNMENT

$$D(\lambda) := \underset{\pi}{\text{maximize}} \quad L(\pi; \lambda)$$

★ explicit maximizer $\pi^\star(\lambda)$

$$\pi^\star(y | x) = \pi_{\text{ref}}(y | x) e^{r(x, y) / \mathbf{1}^\top \lambda} \frac{1}{Z(x)}$$

$$\pi_{\text{ref}}(y | x) := \prod_{k=0}^K (\pi_k(y | x))^{\lambda_k / \mathbf{1}^\top \lambda} \frac{1}{Z_{\text{ref}}(x)}$$

Optimal dualization

■ EXPLICIT DUAL FUNCTION

$$D(\lambda) = \mathbf{1}^\top \lambda \mathbb{E}_x \left[\log \mathbb{E}_{y \sim \pi_{\text{ref}}(\cdot | x)} \left[\exp \left(\frac{r(x, y) + \lambda^\top \epsilon}{\mathbf{1}^\top \lambda} \right) \right] \right] - \mathbf{1}^\top \lambda \mathbb{E}_x [\log Z_{\text{ref}}(x)]$$

$$\pi_{\text{ref}}(y | x) := \prod_{k=0}^K (\pi_k(y | x))^{\lambda_k / \mathbf{1}^\top \lambda} \frac{1}{Z_{\text{ref}}(x)}$$

Independent of π , smooth, and convexity

$\implies ??$ easy to find λ^*

■ SAMPLES FROM REFERENCE MODEL

$$\log \pi_{\text{ref}}(y \mid x) = \sum_{k=0}^K \frac{\lambda_k}{\mathbf{1}^\top \boldsymbol{\lambda}} \log \pi_k(y \mid x)$$

Soup model

■ NORMALIZATION CONSTANT

$$\mathbf{1}^\top \boldsymbol{\lambda} \log Z_{\text{ref}}(x) \approx \prod_{k=0}^K (\pi_k(\textcolor{red}{y}_{\text{max}} \mid x))^{\lambda_k} = e^{\sum_{k=0}^K \lambda_k \log \pi_k(\textcolor{red}{y}_{\text{max}} \mid x)}$$

Max-approximation

Dualization approach

■ NON-ITERATIVE CONSTRAINED ALIGNMENT METHOD

stage 1: find an optimal dual $\lambda^* \in \operatorname{argmax}_{\lambda \geq 0} D(\lambda)$

stage 2: solve the primal problem $\pi^* = \operatorname{maximize}_{\pi} L(\pi, \lambda^*)$

■ PRACTICAL IMPLEMENTATIONS

model-based setting

model-based dual

pseudo-preference optimization

preference-based setting

preference-based dual

pseudo-preference optimization

Summary

■ MOTIVATION

Carbon footprint, generalization, personalization

■ EXAMPLES

Model merging, model soup

■ REGULARIZATION-BASED APPROACH

Mixture of regularizations, personalized regularization, worst-case regularization

■ CONSTRAINED APPROACH

Multiple constraints, optimal dualization